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HYGEIA 2.0

Smart Glove and Container for Automated Patient Monitoring and Medicine Dispensing

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INTRODUCTION

Elderly people often experience serious difficulties in communication and mobility, which makes them highly dependent on caregivers for daily activities, medication intake, and emergency support. In many care facilities, a small number of staff must look after a large number of patients, increasing the risk of delayed or incorrect medication. Even a short delay of a few minutes can become life-threatening. With the rapidly growing aging population, there is a strong demand for smart, reliable, and scalable healthcare solutions that can improve patient safety, independence, and quality of care.

OBJECTIVE

This project aims to develop an integrated IoT-based elderly care system that provides:

- Continuous health monitoring using wearable sensors for heart rate, SpO₂, and body temperature.
- Gesture-based, non-verbal communication using tactile switches.
- Automated and accurate medicine dispensing at scheduled times.
- Remote monitoring through a mobile application with real-time alerts.

Main Outcome: Reduce caregiver workload while improving patient safety and independence.

RESEARCH GAP

Most existing elderly care solutions focus only on a single function such as health monitoring, emergency alerts, or medication reminders. Many systems depend on short-range technologies like Bluetooth (10–15 m), which are not reliable for safety-critical applications, and several lack proper clinical validation. At present, there is no widely adopted, integrated solution that combines gesture-based communication, continuous vital sign monitoring, and automated medicine dispensing using a reliable long-range communication method with centralized remote monitoring.

METHODOLOGY

The system integrates two primary hardware components: a smart wearable glove and an automated medicine dispensing unit.

Smart Glove:

- MAX30100 sensor for heart rate and SpO₂
- DS18B20 digital thermometer for temperature
- Eight tactile switches for gesture-based communication
- Vibration motor for haptic feedback

Dispensing Unit:

- 24-compartment rotating wheel
- ESP32-controlled stepper motor with 1:24 gear ratio
- Moisture-controlled drying chamber with silica gel
- Precise 15° rotation intervals for scheduled dispensing

Communication & Software:

- Dual ESP32 microcontrollers using ESP-NOW protocol for reliable inter-device communication (replacing 433 MHz RF)
- Blynk mobile application for remote monitoring, global connectivity, and real-time push notifications

RESULTS/FINDINGS

Clinical validation across five healthy subjects (ages 21–24) demonstrated:

- Temperature: 99.8% accuracy (±0.15 °C)
- SpO₂: 99.5% accuracy (±1.2%)
- Heart Rate: 99.1% accuracy (±3.4 BPM)

PROTOTYPE TESTING OVER 150 CYCLES ACHIEVED:

- Medication adherence: 96% (baseline 67% → +43% improvement)
- Message transmission success: 99.2%
- Dispensing accuracy: Within ±2 seconds of schedule
- ESP-NOW reliability: 100% packet delivery vs 84.6% with previous RF approach

LIMITATIONS & FUTURE WORK

Future development is planned over 18 months in three stages:

PHASE 1:

- Add ESP32-CAM for fall detection
- Implement automatic water dispensing
- Improve battery life to more than 48 hours

PHASE 2:

- Introduce LSTM-based predictive health analysis
- Expand the system to monitor more than 10 patients from one dashboard
- Work towards FDA Class II medical device certification

PHASE 3:

- Implement HL7 FHIR standard for hospital record system integration
- Add voice command support
- Conduct a commercial pilot deployment in three care facilities (target: 50 patients)

DISCUSSION

The dual-ESP32 system with the ESP-NOW protocol successfully satisfies the reliability requirements of safety-critical healthcare applications. The use of the Blynk platform extends the operating range from a local 15 m system to global access. The eight-button glove design provides a good balance between ease of use and functionality for long-term daily use. The significant improvement in medication adherence (67% to 96%) and the very low alert delay clearly show the system’s practical value in real elderly care environments.

CONCLUSION

HYGEIA presents a reliable, accurate, and scalable solution for modern elderly care. By combining health monitoring, gesture-based communication, and automated medicine dispensing into a single system, it greatly improves medication compliance, communication reliability, and overall patient safety while also reducing the workload of caregivers.

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PHOTOS

